Detection, Not Delineation: Diagnosing Panoptic Quality on Solar Filaments

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Abstract—Panoptic Quality (PQ) is the product of two factors, and a single PQ number cannot say which one is limiting. We report a segmentation system for solar filaments in H-alpha observations that reaches PQ 0.4406 pooled out-of-fold, and use the decomposition to show that the remaining error is almost entirely a detection failure rather than a boundary-precision failure. Segmentation Quality (SQ) is pinned at 0.68 across four architectures spanning convolutional and transformer encoders, while Recognition Quality (RQ) is what moves. The full Dice and IoU distributions confirm the mechanism: matched instances reach a mean Dice of 0.804, but 16.7% of ground-truth filaments score below 0.10, i.e. are missed outright. At the metric's own matching threshold there is no fragmentation and no merging, ruling out a class of post-processing fixes that our own earlier analysis had assumed were relevant. Two substantial interventions—tripling the training data via cross-instrument label transfer, and adding a transformer to the ensemble—both failed, for identifiable reasons: the first teaches boundary imprecision because the transferred labels describe a different observation of an evolving object, and the second adds no error diversity because the two models' predictions correlate at 0.913.

Index Terms—instance segmentation, panoptic quality, solar filaments, error analysis, ensemble diversity

I. Introduction

Solar filaments are dense clouds of plasma suspended above photospheric neutral lines. They are the source region of coronal mass ejections, flares and solar energetic particle events, which makes segmenting them a prerequisite for space weather forecasting. The MAGFiLO 2026 challenge asks for pixel-accurate instance masks on 2048×2048 H-alpha images, scored by Panoptic Quality.

PQ is defined as

$$\mathrm{PQ} = \underbrace{\frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}|}}_{\mathrm{SQ}} \cdot \underbrace{\frac{|\mathrm{TP}|}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|}}_{\mathrm{RQ}}, \quad (1)$$

where TP are the greedy one-to-one matches at IoU > 0.5. The factorisation is the useful part. SQ measures how well a matched instance is delineated; RQ measures how many instances were found at all. A system can reach a given PQ by being precise about few objects or sloppy about many, and the two failures call for completely different remedies.

This paper is organised around that distinction. Our contribution is not a novel architecture—the segmentation network is a stock ConvNeXt-Small UNet—but a diagnosis. We show that:

1) SQ is invariant across four architectures, so boundary precision is not the bottleneck and further architecture search is not the lever.
2) The error is dominated by missed instances: 16.7% of ground-truth filaments have best-overlap IoU below 0.10.
3) At IoU > 0.5 there is no fragmentation and no merging, so instance-splitting post-processing has nothing to act on.
4) Two well-motivated attempts to improve the system—more data and more architectural diversity—failed for identifiable reasons.

The last point is deliberate. A negative result that is measured properly constrains the next attempt; an unmeasured claim does not.

II. Data

The training set contains 707 unique images and 1,154 image entries, because each image carries between one and three independent annotator copies (411 with one, 145 with two, 151 with three), giving 8,199 instance annotations. The test set has 180 images. Images are 2048×2048 8-bit grayscale.

Filaments are small: the median filament pixel fraction is 0.31% and the median instance area is 1,228 px. This imbalance is why a pixel-wise loss alone is insufficient and why the metric's instance-level accounting matters.

Every annotation includes a spine polyline, the filament centreline. We did not use it: it is ground-truth metadata, and the competition rules forbid training on ground-truth metadata beyond the provided annotations.

a) Splitting: Because multiple annotator copies share the same underlying image, a naive random split leaks the same pixels into train and validation. We split by ISO week (343 groups) into five folds, so every copy of an image inherits that image's fold. All reported numbers are pooled out-of-fold: each file is scored by the fold that did not train on it, and each prediction is scored against every annotator copy of that file, matching the official protocol.

III. Method

A. Preprocessing

Ground-based H-alpha observations have a strong radial intensity gradient from limb darkening and atmospheric path length. We estimate the solar disk geometry (centre

TABLE I
Pooled out-of-fold results. SQ and RQ are the factors of Eq. 1. The last two rows are interventions that were rejected.

Configuration	PQ	SQ	RQ
Segmentation only	0.4220	0.6765	0.6238
+ candidate re-scoring	0.4406	0.6784	0.6495
+ GONG label transfer	0.4045	0.6667	0.6095
+ PVTv2-B2 ensemble	0.4098	0.6719	0.6099

and radius) from the image itself and divide out a radial background profile fitted on concentric annuli. This must run at full resolution: the profile spans the whole disk, so a 512 px crop cannot supply it. The flattened image is then normalised with dataset statistics ($\mu = 0.4254$, $\sigma = 0.0809$).

B. Segmentation

A ConvNeXt-Small encoder with an ImageNet-pretrained initialisation feeds a UNet decoder of width 128. Because the finest ConvNeXt feature map is at stride 4, an 11 px filament cannot be reconstructed from it alone, so a stride-1 stem convolution on the input is concatenated into the final decoder block. Training uses 512 px patches, two samples per image, batch size 12, AdamW at 3×10^{-4} with cosine decay, and a BCE + Dice loss under bf16 autocast.[1]

C. Candidate re-scoring

Thresholding the probability map at a low value and labelling connected components deliberately over-produces candidates: a candidate rejected later costs nothing, whereas a filament that never becomes a candidate is unrecoverable. Each candidate is described by 17 scalars—area, elongation, length, orientation, mean/max/90th-percentile probability, contrast against a surrounding ring, bounding-box dimensions, fill ratio, border contact, and centroid—and a gradient-boosted model predicts its probability of being a true positive. Candidates below $\tau = 0.35$ are discarded.

This stage is where the gain comes from, and it is a ranking problem rather than a segmentation problem: the candidate set already contains the right answer, and the scorer's job is to identify which candidates those are.

IV. Results

A. Overall

Table I reports the four configurations we evaluated. The segmentation baseline reaches PQ 0.4220; candidate re-scoring raises it to 0.4406, a paired gain of +0.0154 with a 95% bootstrap CI of $[+0.0109, +0.0199]$ over 2,000 resamples of the 707 files. The public leaderboard score is 0.38.

[1]fp16 diverged to NaN on this architecture; bf16 has fp32's exponent range and does not.

TABLE II
IoU and Dice over three populations. "Per ground-truth" scores missed instances as 0; "per prediction" scores false positives as 0.

Population	n	IoU	Dice	Dice ≥ 0.75
Matched pairs	5,259	0.677	0.804	77.5%
Per ground-truth	8,199	0.501	0.614	49.7%
Per prediction	8,662	0.481	0.591	47.0%

B. SQ does not move

The most informative column of Table I is SQ. Across four architectures—resnet34, DeepCrack with a decoder, ConvNeXt-Small, and PVTv2-B2—SQ stays within 0.667–0.678. Re-scoring improves RQ by +0.026 and SQ by only +0.002.

This is a strong constraint. If boundary precision were the bottleneck, a transformer with global receptive field should have moved SQ. It did not. The network is already delineating filaments about as well as this supervision allows, and the remaining error lies elsewhere.

C. The error is missed instances

To locate it, we measured the full IoU and Dice distributions over all 707 out-of-fold files with the shipped post-processing. Table II reports three populations, because a single histogram is ambiguous: quoting only matched pairs flatters the system by hiding misses, while quoting only per-object values hides that boundaries are good when the instance is found.

The gap between the first two rows is the finding. When an instance is matched, Dice averages 0.804 and 77.5% of matches exceed 0.75: the boundaries are good. Across all ground-truth instances the mean falls to 0.614, because 16.7% of filaments score below 0.10—they were missed entirely. Fig. 1 shows the same thing as a distribution.

Recall is 0.641 and precision 0.607. Both are mediocre, but the asymmetry matters: a missed filament costs 0.5 in the PQ denominator and is unrecoverable, whereas a false positive also costs 0.5 but at least had a chance of being right. The system's dominant failure is not seeing filaments.

D. No fragmentation, no merging

PQ's denominator charges $\frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|$, so splitting one filament into three costs almost as much as missing it. This motivated a fragment-merging stage in our original plan, based on an early analysis that attributed 58.5% of false positives to fragments.

That analysis was wrong. Measuring the false-positive anatomy over all 707 files gives 47.3% background, 51.0% unmatched ground truth, and 1.8% fragments. The oracle gain from removing every fragment is +0.0015 PQ.

Table III shows why. At the metric's own threshold of IoU > 0.5 there is no fragmentation and no merging at all: every overlap is a clean 1:1 or nothing. Multi-relations appear only at loose overlaps, where a prediction grazes a neighbouring filament without competing for the match.

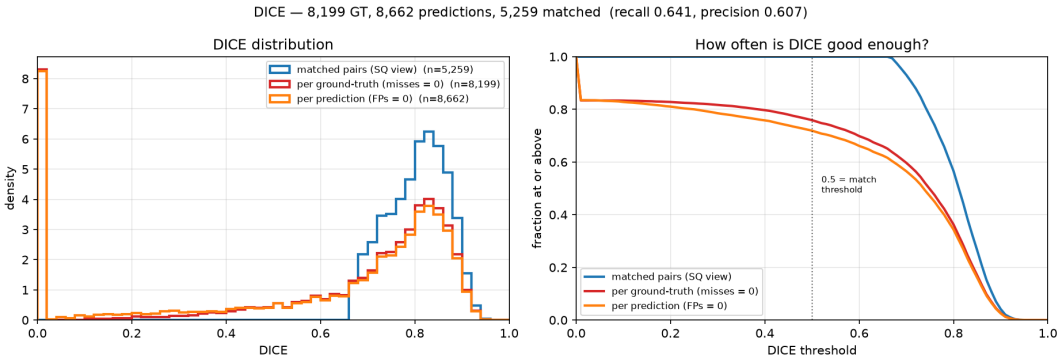


Fig. 1. Dice over three populations. The matched-pair curve (blue) sits well above the per-object curves (red, orange); the mass at zero in the latter is missed filaments and false positives. Left: density. Right: the fraction of each population reaching at least a given Dice, which is the form in which the 16.7% miss rate is directly readable.

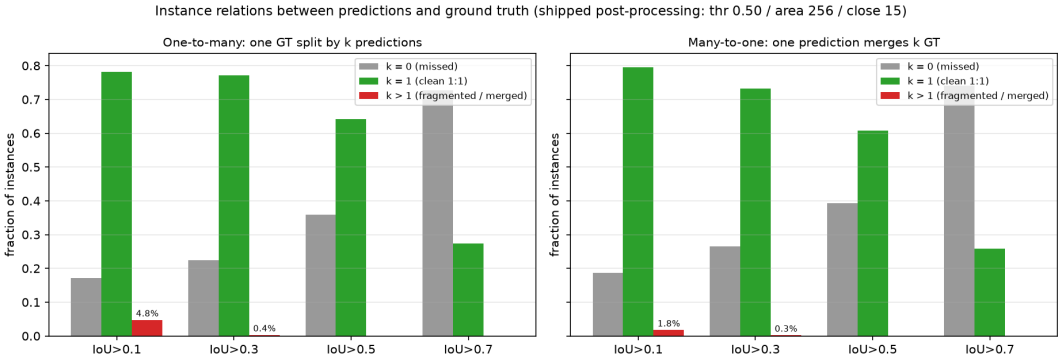


Fig. 2. Relation histogram. The $k > 1$ bars (red) are the relations PQ punishes; they are absent at the matching threshold of 0.5 and appear only at loose overlaps, where a prediction grazes a neighbour without competing for the match.

TABLE III
Instance relations. At the matching threshold of 0.5 the multi-relations vanish entirely.

Relation	IoU > 0.1	IoU > 0.5
One GT split by $k > 1$ predictions	4.8%	0.0%
One prediction merging $k > 1$ GT	1.8%	0.0%

V. Two interventions that failed

A. More data, from the wrong observation

The competition images are GONG H-alpha frames, so annotations can in principle be transferred from nearby frames in the public archive. We verified that the competition images are the GONG hag JPEGs (correlation 0.997, median absolute difference 1 count, best alignment shift $(0, 0)$), which made transfer feasible without radiometric correction.

Transferring labels from nearby frames produced 3.1× the training data and a PQ change of −0.0150. The per-year breakdown explains it: gains appeared exactly where the extra frames were concentrated (2021–22, +0.0109), and losses appeared in years where transferred labels came from a different observation (2011: −0.0475; 2016: −0.0426; 2017: −0.0389). SQ fell from 0.6750 to 0.6667.

The mechanism is physical. Solar filaments evolve over hours. A polygon drawn on a frame six hours earlier describes a different object, and training on it teaches the model to be imprecise about boundaries. More data of the wrong kind is worse than less data of the right kind.

B. Architectural diversity without error diversity

Ensembles help when members make uncorrelated errors. Our five ConvNeXt folds share an architecture, a preprocessing pipeline and a loss, so averaging them mostly reduces variance. A pyramid vision transformer (PVTv2-B2) was expected to fail differently: self-attention gives every output position a path to every input position, so a filament is seen as a whole shape rather than through stacked local receptive fields.

PVTv2 scored 0.3906 solo against ConvNeXt's 0.4177, which is expected for a diversity member. But averaging the two gave −0.0113 (CI $[-0.0239, +0.0001]$). The mechanism is one number: the two models' predictions correlate at **0.913**.

Two predictors that agree on 91% of pixels cannot cancel each other's mistakes. The average simply lands between them, and below the better one. Architectural difference is not the same thing as error diversity—the same lesson

our earlier attempt to replace the scalar scorer with a CNN over image crops produced, where the CNN reached a correlation of 0.507 with true candidate quality against the gradient-boosted model's 0.575.

VI. Where the remaining error is

Scoring every candidate with its true label gives an oracle ceiling of PQ 0.5028, i.e. +0.0808 above the segmentation baseline. The learned scorer captures +0.0186 of that (23%), leaving +0.0622 unclaimed.

This is the largest known opportunity in the system, and it is a ranking problem rather than a segmentation problem. The candidates already contain the right answer; the scorer cannot tell which ones they are. Given that the error is dominated by missed instances, the productive direction is a scorer that is better at recall—recovering candidates that the current model rejects confidently—rather than one that is better at precision.

VII. Reproducibility

Code, pinned requirements and a pipeline notebook are in the public repository. Weights are published as release assets and fetched by a script. Inference requires no network access: the checkpoint supplies every weight, so the pretrained backbone is not downloaded. All reported numbers are produced by scripts in the repository from the out-of-fold predictions, and the report itself is generated from a machine-readable summary so that no figure can drift from the measurement that produced it.

VIII. Discussion

A. Why the decomposition matters

It is tempting to treat PQ as a leaderboard number and optimise it directly. The factorisation in Eq. 1 makes that a mistake. Our four configurations span a PQ range of 0.036, but SQ varies by only 0.012 across all of them while RQ varies by 0.040. Any intervention that does not change RQ is working against a ceiling it cannot move.

This also explains a pattern that would otherwise look like a plateau. Four architectures—a ResNet, a DeepCrack variant with an explicit decoder, a ConvNeXt, and a transformer—all land within 0.011 SQ. That is not a coincidence of tuning. It is what happens when the supervision, not the architecture, sets boundary precision: the annotators disagree about where a filament edge is, and the model has learned the average of that disagreement.

B. The cost of a missed instance

The asymmetry between precision (0.607) and recall (0.641) is small in isolation, but the two errors are not equally recoverable. A false positive is a candidate the scorer accepted; a missed filament is one it rejected, and no downstream stage can bring it back. Since the candidate set already contains +0.0808 PQ of headroom, the missed filaments are largely present as candidates and being discarded.

That reframes the problem as one of scorer calibration rather than candidate generation. A scorer trained to maximise average precision will happily discard low-confidence candidates that are nonetheless true positives, because they contribute little to the ranking metric. PQ does not care about ranking—it cares about the count of matched instances. The objective mismatch is a plausible explanation for why only 23% of the oracle gap was captured.

C. What would not have helped

Both failed interventions were well-motivated: label transfer addressed data scarcity, ensemble diversity addressed variance. Neither addressed detection, and neither could have moved the number that was limiting. The general lesson is that a diagnosis has to precede an intervention. Here it arrived late—the fragment-merging stage was designed and nearly built before anyone measured how many fragments existed, at which point the answer was 1.8% of false positives and 0.0% of matches at the operating threshold.

D. Limitations

All local numbers are pooled out-of-fold on 707 training images; the public leaderboard is a two-decimal readout on roughly half the test set. The local-to-public gap is 0.0586 and is not constant across component types, so it is not a fixed offset to subtract. The test set is also shifted—11.7% of test files are from 2021–22 against 5.7% of training files—and since label transfer showed sensitivity to observation epoch, this is a plausible contributor we have not isolated.

IX. Conclusion

A single PQ number hides which of its two factors is limiting. Decomposing it showed that our system's boundary precision is saturated—SQ is invariant across four architectures—and that the remaining error is missed instances, not badly drawn ones. That reframing is what made the two failed interventions interpretable: neither addressed detection, and neither could have helped.

The immediate next step is a candidate scorer optimised for recall against the +0.0622 oracle gap, rather than further architecture search.

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